LECTURES

Cristian Giardinà - Integrable models of heat conduction

This series of lectures aims to illustrate the connection between integrability and stochastic models of heat transport. This connection is rooted in the description of the generator of a stochastic process as the quantum Hamiltonian of a spin system. As we shall see, this connection is useful to better understand existing models, as well as to introduce new exactly solvable models.

In the first lecture, we introduce the notion of duality in the context of Markov processes and its relation to quantum spin chains, symmetry, and representation theory of Lie algebras.
In the second lecture, we discuss a few well-known heat conduction models, such as the Kipnis-Marchioro-Presutti (KMP) model, and the use of duality to characterize its non-equilibrium steady state.
In the third lecture, we present the recently introduced integrable version of the KMP model (so-called harmonic model) [1] and prove large deviations for the energy density [2]. If time permits, we will also discuss dynamical large deviations (ongoing work [3])

References:
[1] R. Frassek, C. Giardinà, J. Kurchan, *Non-compact quantum spin chains as integrable stochastic particle processes*, J. Stat. Phys. 180, 135–171 (2020); R. Frassek, C. Giardinà, *Exact solution of an integrable non-equilibrium particle system*, J. Math. Phys. 63, 103301 (2022).
[2] G. Carinci, C. Franceschini, R. Frassek, C. Giardinà, F. Redig, *Large Deviations and Additivity Principle for the Open Harmonic Process*, Comm. Math. Phys. 406 (2025).
[3] C. Giardinà, T. Sasamoto; *Large spin large deviations for interacting particle systems*; C. Giardinà, K. Mallick, T. Sasamoto, H. Suda, *Exact solution of discrete macroscopic fluctuation theory for an integrable spin system*, in preparation.

Lorenzo Piroli - Exactly solvable models of random quantum circuits
n/a

SEMINARS

Davide Lai - Clustering and the Five-Point Function

The hexagonalisation technique arose in the context of N=4 SYM theory to address the tessellation of the effective world-sheet describing three-point functions of single trace operators.
Despite its apparent technical difficulty, exploiting the knowledge of all its ingredients at finite 't Hooft coupling one can perform some non-trivial limits, like the strong-coupling one. In this case, a phenomenon called "clustering" allows the complete resummation of finite-size effects leading to the semi-classical string answer. I will introduce the general background and then discuss the generalization to some particular class of five-point functions, where the clustering phenomenon becomes more involved, involving cross-ratios and some

analytic continuation of the ingredients to unusual kinematic regimes. In the end, I will speculate about some tentative directions that one could take.

Filiberto Ares - Entanglement asymmetry and the quantum Mpemba effect

The Mpemba effect is the counterintuitive and controversial phenomenon that hot water cools faster than cold one, or in more formal words, the more a system is out of equilibrium, the faster it relaxes. We introduce an analogous effect in extended quantum systems, in which a symmetry explicitly broken by the initial state is dynamically restored by the time evolution after a quantum quench. Unexpectedly, we find that the more the symmetry is initially broken, the faster is restored. To study this phenomenon, we borrow methods from the theory of entanglement in many-body systems and we introduce a quantity, dubbed entanglement asymmetry, which measures how much a symmetry is broken in a subsystem. In contrast to the classical case, we can establish the criteria for its occurrence in arbitrary integrable quantum systems and provide direct experimental evidence using a trapped-ion quantum simulator.

Niels Benedikter - Derivation of the Luttinger Model from the Interacting Fermi Gas

The Luttinger model is famous as a model of interacting fermionic quantum particles in one space dimension due to the fact that it can be exactly solved using the bosonization method discovered by Mattis and Lieb in 1965. This method however is very specific to the model with its linear dispersion relation and its particular form of interaction. I will discuss how the Luttinger model however arises in a mean-field limit of a general fermionic quantum many-body model which is not exactly solvable.

Francesco Ravanini - Hagedorn singularity in exact U_q(su(2)) S-matrix theories with arbitrary spins

Generalizing the quantum sine-Gordon and sausage models, we construct exact S-matrices for higher spin representations with quantum U_q(su(2)) symmetry, which satisfy unitarity, crossing-symmetry, and the Yang-Baxter equations with minimality assumption, i.e. without any unnecessary CDD factor. The deformation parameter q is related to a coupling constant. Based on these S-matrices, we derive the thermodynamic Bethe ansatz equations for q a root of unity in terms of a universal kernel where the nodes are connected by graphs of non-Dynkin type. We solve these equations numerically to find out Hagedorn-like singularities in the free energies at some critical scales and find a universality in the critical exponents, all near 0.5 for different values of the spin and the coupling constant.

Henrik Juergens - About the structure of the correlation functions of the sl(n+1) invariant vertex model

My talk will be about the structure of the correlation functions of the sl(n+1) invariant vertex model and it's corresponding spin chain. In the case n=1, i.e. the XXX and XXZ spin chain, there is a surprising result due to a series of papers of Jimbo et al. which ultimately allows the exact calculation of correlation functions both in the rational and trigonometric case through a 'hidden' fermionic structure. However, although this structure seems quite universal for calculating correlation functions for finite and infinite size, zero and finite temperature (and even form factors), there is to this day no clear understanding if or how this structure will generalise when considering higher rank symmetries. I will present an ansatz for generalising the initial construction of recursion relations for the correlation functions in the rational case (Jimbo et al. 2005) which has finally been published in April this year. As the topic is quite complicated, I will give a short overview about correlation functions, explain the 2005 construction and then focus on discussing the difficulties that come up when generalising. Moreover, I will conclude by interpreting the results in terms of so called snake modules - certain (prime) irreducible representations of the Yangian.

Andrea Appel - n/a

Davide Fioravanti - n/a

Marco Meineri - n/a

Lenart Zadnik - n/a

Francesco Buccheri - Orbital angular momentum control with Weyl semimetals

Twisted light beams [1] can carry large amounts of information over large distances, as well as selectively induce atomic and molecular transitions. Because of this, they hold significant promise for quantum computation and sensing technologies. In this talk, I will argue that Weyl semimetals, topological materials in which valence and conduction bands touch in a finite number of points [2], can be used to effectively manipulate twisted light. I will show that, because of the presence of an axionic term in the emergent electrodynamics [4,5], their surface plasmon nonreciprocity can be used to efficiently control the propagation of light in a cylindrical waveguide. As a consequence, a conical tip will focus light with given sign of the angular momentum only [5].

[1] L. Allen, M. W. Beijersbergen, R. J. C. Spreeuw, and J. P. Woerdman, "Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes" , Phys. Rev. A 45, 8185
[2] N. P. Armitage, E. J. Mele, and A. Vishwanath, "Weyl and Dirac semimetals in three-dimensional solids", Rev. Mod. Phys. 90, 015001 (2018);
[3] A. A. Zyuzin and A. A. Burkov, "Topological response in Weyl semimetals and the chiral anomaly", Phys. Rev. B 86, 115133 (2012);
[4] M. M. Vazifeh and M. Franz, Electromagnetic response of Weyl semimetals, Phys. Rev. Lett. 111, 027201 (2013);
[5] M. Peluso, A. De Martino, R. Egger, and F. Buccheri, "Nonreciprocal Weyl semimetal waveguide", arXiv:2410.01503 (2024) & "Selective plasmon nanofocusing with Weyl semimetals", in preparation.

Saverio Rota - n/a